

Notes: Hertzberg, Liberti, and Paravisini (JF, 2011)

Public Information and Coordination: Evidence from a Credit Registry Expansion

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1 Big picture

- **Question.** Does making credit information public change lenders' behavior because lenders want to coordinate with each other, rather than only because lenders learn new borrower fundamentals?
- **Setting.** The paper studies the 1998 expansion of Argentina's public credit registry. Before the reform, borrower-level credit information was shared only for borrowers above a \$200,000 total-debt threshold, or for borrowers with bad credit histories. The reform effectively lowered the information-sharing threshold to \$50.
- **Natural experiment.** The reform was announced before it was fully implemented. This creates an interim period in which a lender knows that its own private rating will become public but has not yet received other lenders' private ratings.
- **Core empirical idea.** If a bank already knows its own rating of the borrower, the announcement alone should not change its lending unless it anticipates the reactions of other lenders. A change in lending before new information arrives is therefore evidence of a coordination channel.
- **Treatment and control.** Treatment firms are just below the pre-reform threshold, with total debt between \$150,000 and \$200,000. Control firms are just above it, with total debt between \$200,000 and \$250,000. The main estimates compare changes around the reform for these two groups.
- **Main finding.** Banks that had assigned a bad rating to a multi-lender borrower cut lending immediately after the expansion announcement. The paper estimates a 19.8% decline in interim-period lending by banks that had rated the borrower as a poor risk.
- **Coordination evidence.** The effect appears only for firms with multiple lenders. It does not appear for firms with a single lender, where inter-lender coordination incentives are absent.
- **Financial distress.** The reform announcement raises the monthly default hazard for multi-lender firms with at least one bad rating by 5.6 percentage points during the two-month interim period.
- **Long-run adjustment.** After information becomes public, firms concentrate debt more with their main lender. This is consistent with firms and banks trying to reduce future coordination problems.
- **Economic takeaway.** Public information can have a multiplier effect: news matters not only because it changes beliefs about fundamentals, but also because it changes beliefs about what other creditors will do.

2 Incremental contribution of the paper

- **Direct evidence on creditor coordination.** The paper moves beyond indirect evidence of creditor runs by isolating a setting in which a lender changes behavior even though it has not learned new information about the borrower.
- **A clean publicity-multiplier test.** The interim period separates the effect of making a signal public from the effect of receiving new signals. This is the key contribution relative to standard information-sharing studies.

- **Natural experiment from registry technology.** The expansion was driven by the Central Bank’s adoption of CD-ROM distribution, which reduced the cost of disseminating information. This institutional change creates plausibly exogenous variation around the \$200,000 threshold.
- **Difference-in-differences around a threshold.** By comparing borrowers just below and just above the old registry threshold, the paper controls for aggregate shocks to credit markets while focusing on economically similar firms.
- **Lender-level mechanism.** The paper distinguishes banks that possessed bad news from banks that did not. The strongest interim response comes from the banks that already held the bad signal, which supports the anticipation channel.
- **Firm distress outcome.** The paper does not stop at credit quantities. It shows that the public-information shock also increases default hazards, linking coordination to real financial distress.
- **Long-run equilibrium response.** The paper documents an endogenous organizational response in credit structure: affected firms concentrate borrowing more heavily with fewer lenders after information becomes public.
- **Policy contribution.** The paper highlights a cost of transparency in credit markets. Public information can improve learning, but it can also intensify strategic complementarities among creditors when borrowers are near distress.

3 Institutional setting and data

3.1 Argentina’s public credit registry before 1998

- Argentina’s public credit registry was established in 1991 and collected monthly borrower-level information from financial institutions.
- Banks reported total outstanding debt, collateral pledged, and a credit-risk rating for each borrower.
- Ratings range from 1 to 5. A rating of 1 is the safest category; 2 indicates potential repayment problems; 3 indicates high potential default risk or serious arrears/refinancing needs; ratings 4 and 5 are treated as default in the paper.
- Beginning in 1995, financial institutions could access full current credit records only for certain borrowers. The relevant rule shared information for borrowers with total debt above \$200,000 at any time during the prior 12 months or borrowers with worse credit histories.
- Thus, before the reform, many smaller borrowers’ information was reported to the Central Bank but was not shared with other lenders.

Interpretation: The data contain both the information that banks reported and the information-sharing rules that determined whether those reports were visible to other lenders. This is what makes the registry expansion useful for identification.

3.2 The 1998 registry expansion

- In 1998, the Central Bank adopted CD-ROM distribution for registry data.

- The lower cost of information distribution made the \$200,000 threshold obsolete, and information became available for virtually all borrowers with total debt above \$50.
- The expansion was retroactive to January 1998, but it was not announced until April and was not fully implemented until July.
- This timing creates three periods: a pre-announcement period, an interim period after announcement but before information release, and a postexpansion period after information was released.

Interpretation: The interim period is central. During this period, a bank knows its own past reports will become public, but it has not yet learned other banks' reports. Any lending response by that same bank is therefore not driven by new registry information arriving to it.

3.3 Treatment and control samples

- The treatment group consists of firms with total debt between \$150,000 and \$200,000 during January–March 1998.
- The control group consists of firms with total debt between \$200,000 and \$250,000 during the same pre-announcement period.
- Control borrowers were already covered by information sharing before the expansion, while treatment borrowers became newly covered by the expansion.
- The main sample excludes firms with ratings worse than 2 before the announcement, because the registry expansion mainly affected firms with ratings of 1 or 2.
- The key coordination tests focus on firms with at least one rating of 2 before the expansion, because a rating of 2 is bad news relative to the very high prior probability of rating 1 in this sample.

Interpretation: The design is a local comparison around the registry threshold. Firms just below and just above the cutoff are not identical, but the threshold provides a plausible counterfactual for aggregate shocks.

4 Theoretical framework

4.1 Coordination environment

Two banks lend to the same firm. Bank i lends L_i and receives a gross interest rate R if the loan is repaid. The probability that bank i is repaid is increasing in the creditworthiness of its own loan and in the amount lent by the other bank:

$$\Pr(\text{repay } i) = \theta_i L_j. \quad (1)$$

The bank's objective is

$$\max_{L_i \geq 0} \theta_i E_i(L_j) L_i R - \frac{1}{\delta} L_i^\delta, \quad (2)$$

where $\delta > 2$.

- θ_i is bank i 's private assessment of the creditworthiness of its loan.

- $E_i(L_j)$ is bank i 's expectation of how much the other bank will lend.
- The complementarity captures the creditor-coordination problem: if one lender withdraws, the borrower's repayment prospects for other lenders also decline.

Interpretation: The key feature is that each lender's payoff depends on the other lender's action. Public information matters because it helps a lender predict the other lender's response.

4.2 Publicity multiplier

The first-order condition is

$$\theta_i E_i(L_j)R - L_i^{\delta-1} = 0. \quad (3)$$

Without information sharing, bank i 's private signal does not affect the other bank's expected action, so the elasticity of lending with respect to θ_i is

$$\eta^{NS} = \frac{1}{\delta - 1}. \quad (4)$$

With information sharing, bank i 's signal also changes bank j 's lending, which feeds back into bank i 's own lending. The elasticity becomes

$$\eta^S = \frac{\frac{1}{\delta-1}}{1 - \left(\frac{1}{\delta-1}\right)^2}. \quad (5)$$

The publicity multiplier is therefore

$$\Delta\eta = \eta^S - \eta^{NS} > 0. \quad (6)$$

Interpretation: Public information has a larger effect than private information because it changes both beliefs about fundamentals and beliefs about other lenders' actions. This is the paper's key theoretical mechanism.

5 Empirical design and specifications

5.1 Main difference-in-differences specification

The main firm-level specification is

$$\ln(Debt_{it}) = \alpha_i + \xi_t + \gamma_{Interim}Treat_i \times Interim_t + \gamma_{Post}Treat_i \times Post_t + \varepsilon_{it}. \quad (7)$$

- $Treat_i$ equals one for firms newly affected by the registry expansion.
- $Interim_t$ equals one after the announcement but before information release.
- $Post_t$ equals one after actual registry expansion.
- Firm fixed effects absorb time-invariant borrower differences.
- Month dummies absorb aggregate credit-market shocks.
- Standard errors are clustered at the firm level.

Interpretation: The interim coefficient is the key coordination estimate. It measures how debt changes when the bank knows information will become public but before the bank receives new information from other lenders.

5.2 Lender-level triple-difference specification

To compare lending by banks with bad news to lending by banks with good news about the same firm, the paper estimates

$$\ln(Debt_{ijt}) = \alpha_{ij} + \xi_{jt} + Treat_i \times \delta_t \quad (8)$$

$$+ \gamma_{Interim, Bank2} Treat_i \times Interim_t \times BankAssigned2_j \quad (9)$$

$$+ \gamma_{Post, Bank2} Treat_i \times Post_t \times BankAssigned2_j + \varepsilon_{ijt}. \quad (10)$$

- *BankAssigned2_j* equals one for banks that assigned the firm a rating of 2 before the announcement.
- Firm-rating fixed effects compare banks within the same borrower.
- Rating-month fixed effects control for aggregate shocks to banks with good versus bad pre-expansion ratings.
- Treatment-month interactions control for shocks to treated firms.

Interpretation: This specification asks whether banks with bad private information cut lending more than banks with good private information to the same firm when the bad information is about to become public.

5.3 Default hazard specification

The default outcome is estimated using

$$1[Default_{it} = 1 \mid Default_{it-1} = 0] = \xi'_t + \lambda_{Interim} Treat_i \times Interim_t + \lambda_{Post} Treat_i \times Post_t + \zeta_{it}. \quad (11)$$

- The dependent variable equals one when a borrower newly enters default.
- The sample drops firm-months after default occurs.
- The specification asks whether the registry expansion changes the default hazard of treated firms relative to control firms.

Interpretation: If public information triggers lender withdrawal, then the credit contraction should also show up in default risk. The default estimates connect coordination to real borrower distress.

5.4 Identification assumptions and threats

- The key assumption is parallel trends: absent the registry expansion, treatment and control borrowers near the threshold would have followed similar debt paths.
- Figure 2 supports this assumption for the main multi-lender sample because treatment and control debt move similarly before the announcement.
- The paper checks that the effect does not appear for single-lender firms, where the coordination mechanism should be absent.
- Placebo tests shift the reform year to 1999 and shift the cutoff to \$300,000. These placebo exercises do not reproduce the main effects.

- The design cannot rule out every possible unobservable difference near the threshold, but the timing, lender-level evidence, single-lender placebo, and formal placebo tests all point toward the coordination interpretation.

6 Results

6.1 Lending response to bad private information

- For firms with multiple lenders and at least one rating of 2, debt from banks that assigned a rating of 2 falls by 19.8% in the two-month interim period.
- Debt from banks that assigned only a rating of 1 does not fall significantly in the interim period.
- After information is publicly released, banks that had assigned a rating of 1 also reduce lending, consistent with those banks learning the other bank's bad rating.
- Total firm debt falls by 13.5% on average in the postexpansion period for the affected multi-lender sample.

Interpretation: The bad-rating bank cuts before it learns anything new. This is the cleanest evidence that the bank is reacting to the anticipated reaction of other creditors.

6.2 Triple-difference evidence

- Table III shows that lending by banks that assigned rating 2 falls by 21.4% relative to lending by banks that assigned rating 1 to the same firm during the interim period.
- The postexpansion relative decline is 31.9%.
- The result survives a rich set of fixed effects that compare lending within borrower and control for rating-specific aggregate shocks.

Interpretation: This is a within-firm mechanism test. It rules out explanations based only on treatment firms being hit by different aggregate shocks.

6.3 Financial distress

- The default hazard rises by 5.6 percentage points for multi-lender treatment firms with at least one rating of 2 during the two-month interim period.
- The postexpansion default hazard rises by 3.0 percentage points on average.
- The effect is not present for single-lender firms.

Interpretation: The credit contraction is economically important. It is not just a portfolio rebalancing by banks; it increases the probability that firms experience financial distress.

6.4 Placebo tests

- The first placebo assumes the registry expansion occurred in 1999 instead of 1998.
- The second placebo uses a false cutoff of \$300,000 instead of \$200,000.
- Neither placebo produces debt or default effects similar to the main estimates.

Interpretation: The placebo tests reduce the concern that the results mechanically arise from how the sample is selected or from seasonal debt dynamics around the cutoff.

6.5 Long-run effects and debt concentration

- Firms with perfect credit records and multiple lenders experience a modest postexpansion decline in debt, while single-lender firms do not.
- Debt concentration rises after the registry expansion. Debt HHI increases by about 0.0148 in the postexpansion period and by 0.0194 in the second half of the postexpansion year.
- The fraction of debt with the top lender also increases, suggesting that firms concentrate borrowing with their main creditor.

Interpretation: If public information makes coordination failures more likely, firms and lenders have incentives to reduce the number of creditor relationships or to give one lender a larger role.

7 Tables and figures

7.1 Table I: Descriptive Statistics, March 1998 Cross Section

Read:

- Panel A describes the broad analysis sample of firms with ratings of 2 or better before the registry expansion announcement.
- The average borrower has about \$214,600 in debt and a median of two lenders.
- Treatment and control firms are close in many characteristics such as number of lenders, debt concentration, collateral-to-debt ratios, and average risk rating.
- Panel B focuses on the crucial bad-news sample: firms with at least one rating of 2 before the announcement. These firms have slightly more lenders on average and more disagreement across banks' ratings.

Economic message: Table I establishes the local comparison sample and shows that the treatment and control groups near the threshold are broadly similar on observed credit characteristics. It also identifies the smaller subsample where a bank's rating of 2 is the relevant private bad news.

7.2 Figure 1: Borrower Distribution and Characteristics

Read:

- The vertical line marks the \$200,000 pre-reform information-sharing threshold.
- The number of borrowers jumps above the threshold, consistent with the old information-sharing rule shaping the distribution.
- Collateral-to-debt and the fraction with rating 1 look broadly smooth around the cutoff.
- Debt concentration differs around the threshold, which motivates later analysis of endogenous debt concentration.

Economic message: Figure 1 supports the use of a local treatment-control comparison while also foreshadowing the debt-concentration channel: information sharing itself may affect how many lenders a firm uses.

Table I
Descriptive Statistics, March 1998 Cross Section (before Expansion Announcement)

This table presents summary statistics of the firm cross section in our sample of treatment and control firms whose highest (worst) risk rating during the pre-announcement period is a 2. Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Firm debt HHI is the sum of the squared fractions of debt from each lender. We define the lead bank as the lender that provides the largest amount of credit. Risk ratings are assigned by each lender to a firm, and are integers between 1 (best) and 5 (worst). A rating of 1 (2) represents a firm in good standing with no (some) potential repayment problems. For firms with multiple lenders, we report the standard deviation of the ratings assigned by different lenders to the same firm.

Sample	All			Treatment Firms			Control Firms		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
Panel A: Firms with Rating of 2 or Better before the Registry Expansion Announcement									
Firm-level statistics									
Total debt ('000)	214.6	217.0	22.8	178.7	181.2	14.3	223.4	222.1	14.3
Number of lenders	1.88	2.00	1.06	1.89	2.00	1.13	1.88	2.00	1.04
Debt concentration (HHI)	0.84	1.00	0.22	0.83	1.00	0.23	0.85	0.99	0.21
Fraction debt from lead bank	0.88	1.00	0.17	0.88	1.00	0.18	0.89	1.00	0.17
Collateral/Debt	0.61	0.78	0.40	0.57	0.71	0.39	0.61	0.80	0.40
Average risk rating	1.10	1.00	0.31	1.12	1.00	0.32	1.10	1.00	0.31
SD of same firm ratings	0.13	0.00	0.32	0.16	0.00	0.35	0.12	0.00	0.31
No. of observations	1,715	1,715	1,715	337	337	337	1,378	1,378	1,378
Firm-bank relationship level statistics									
Debt ('000)	114.0	114.3	88.0	94.7	93.1	71.1	118.8	119.7	91.1
Risk rating	1.10	1.00	0.37	1.12	1.00	0.39	1.10	1.00	0.36
No. of observations	3,228	3,228	3,228	636	636	636	2,592	2,592	2,592

(continued)

(a) Table I, Panel A: firms with rating of 2 or better

Table I—Continued

Sample	All			Treatment Firms			Control Firms		
	Mean	Median	SD	Mean	Median	SD	Mean	Median	SD
Panel B: Firms with at Least One Rating of 2 before the Registry Expansion Announcement									
Firm-level statistics									
Total debt ('000)	216.0	221.8	25.7	177.0	176.6	13.8	228.0	229.2	13.9
Number of lenders	2.18	2.00	1.21	2.18	2.00	1.38	2.18	2.00	1.16
Debt concentration (HHI)	0.79	0.90	0.24	0.78	0.90	0.24	0.79	0.90	0.24
Fraction debt from lead bank	0.85	0.95	0.19	0.84	0.95	0.18	0.85	0.95	0.19
Collateral/Debt	0.67	0.81	0.35	0.66	0.80	0.35	0.67	0.81	0.35
Average risk rating	1.61	1.50	0.45	1.55	1.53	0.53	1.63	1.50	0.43
SD of same firm ratings	0.53	0.58	0.39	0.46	0.58	0.42	0.55	0.58	0.37
No. of observations	208	208	208	49	49	49	159	159	159
Firm-bank relationship level statistics									
Debt ('000)	99.0	72.0	85.1	81.0	67.1	67.6	104.5	74.7	89.2
Risk rating	1.50	1.00	0.63	1.44	1.00	0.65	1.51	1.00	0.62
No. of observations	454	454	454	107	107	107	347	347	347

(b) Table I, Panel B: firms with at least one rating of 2

7.3 Figure 2: Average Log Total Debt

Read:

- Panel A shows multi-lender firms with at least one rating of 2. Treatment and control debt trends are similar before the announcement.
- After the announcement, treatment-firm debt falls sharply, while control-firm debt does not.
- Panel B shows single-lender firms. Treatment and control debt remain parallel both before and after the announcement.

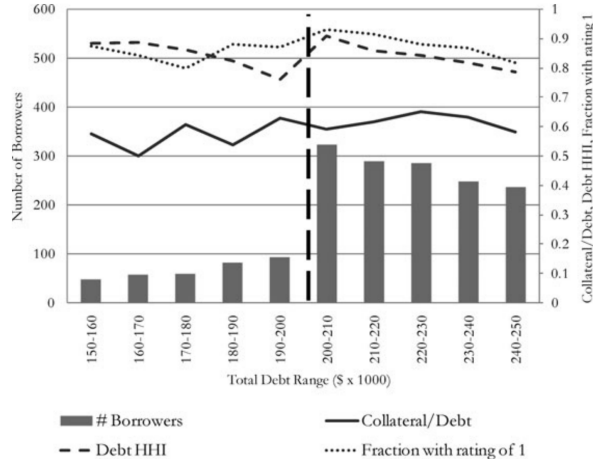
Economic message: Figure 2 is the key visual evidence. The debt drop appears exactly where the coordination channel predicts it should appear: multi-lender borrowers with bad information that is about to become public.

7.4 Table II: Effect of Registry Expansion on Log Debt

Read:

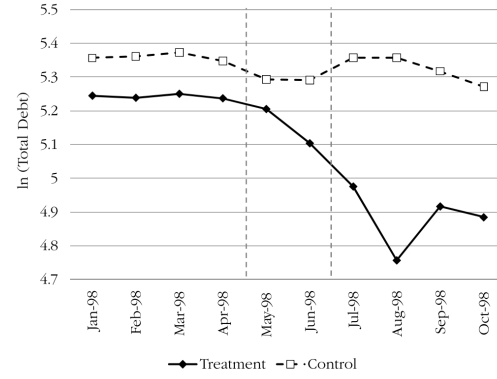
- Column 1 shows that debt from banks that assigned rating 2 falls by 19.8% in the interim period and 35.6% in the postexpansion period.
- Column 2 shows no significant interim decline for banks that assigned rating 1, but a significant 11.5% postexpansion decline.
- Column 3 shows total debt falls by 13.5% in the postexpansion period for multi-lender firms.
- Column 4 shows no meaningful effect for single-lender firms.

Economic message: The banks with bad private information act before any new information arrives. Banks without the bad signal act only after the bad signal becomes public.

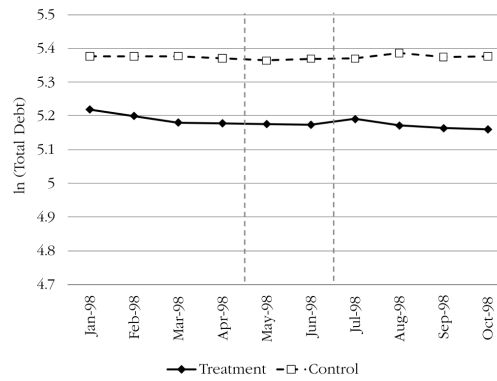


(a) Figure 1: borrower distribution and characteristics

Panel A. Firms with Multiple Lenders before Expansion Announcement



Panel B. Firms with a Single Lender before Expansion Announcement



(b) Figure 2: log debt for treatment and control firms

7.5 Table III: Relative Lending by Bad-News Banks

Read:

- Table III adds a third difference: lending by banks that assigned rating 2 relative to lending by banks that assigned rating 1 to the same firm.
- The interim-period triple-difference coefficient is -0.214, significant at the 5% level.
- The postexpansion triple-difference coefficient is -0.319, also significant.
- The specification includes firm-by-bank/rating fixed effects, bank-by-month dummies, and treatment-by-month dummies.

Economic message: Table III shows that the lending decline is specifically concentrated among banks with bad private information. This strengthens the publicity-multiplier interpretation.

7.6 Table IV: Effect on Default Hazard Rate

Read:

- For multi-lender firms, the default hazard rises by 5.6 percentage points during the interim period.
- The postexpansion default hazard rises by 3.0 percentage points.

- For single-lender firms, the estimated interim effect is close to zero and negative.

Economic message: Publicity-driven credit withdrawal can push firms into distress. This is evidence that the coordination channel has real consequences rather than merely reallocating credit across lenders.

Table II
Effect of Registry Expansion on ln(Debt), Subsample of Firms with at Least One Rating of 2 before Expansion

This table presents the DD effect of the registry expansion announcement (interim period) and public information (postexpansion period) on (log) debt levels, using specification (5) estimated with OLS:

$$\ln(\text{Debt}_{ijt}) = \alpha_i + \beta_1 + \gamma_{\text{Interim}} \text{Treat}_i \times \text{Interim}_t + \gamma_{\text{Post}} \text{Treat}_i \times \text{Post}_t + \epsilon_{ijt}$$

Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Sample: firms whose highest (worst) risk rating during the pre-announcement period is a 2, and with at least one rating of 2 (firms with only good ratings excluded). Columns 1 through 3 are estimated over the subsample of firms with multiple lenders, and column 4 on the subsample with a single lender, before the expansion announcement. Dependent variables: (log) debt of borrower i in month t with the banks that assigned the worst rating (column 1), with the banks that assigned the best rating (column 2), and with all lenders (columns 3 and 4). Right-hand-side variable of interest: interaction between a dummy equal to one if borrower i was in the treatment group (information not shared before registry expansion), and dummies equal to one during the interim and postexpansion periods. Coefficients γ_{Interim} and γ_{Post} are the DD estimates of the effect of the registry expansion announcement and actual expansion on debt. Standard errors are heteroskedasticity robust and clustered at the firm level. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Subsample by # of Lenders before April Dependent Variable	Multiple Lenders			Single Lender ln(Debt _{it})
	ln(Debt from banks w/rating = 2 _{it}) (1)	ln(Debt from Banks w/ Rating = 1 _{it}) (2)	ln (Debt _{it}) (3)	
Effect on debt level: interim period (2 months)	-0.198* (0.102)	-0.058 (0.065)	-0.075 (0.078)	-0.013 (0.131)
Effect on debt level: postexpansion (12 months)	-0.356** (0.148)	-0.115** (0.057)	-0.135*** (0.045)	0.000 (0.180)
Firm fixed effects and month dummies	Yes	Yes	Yes	Yes
No. of observations (firm-month)	2,381	2,368	2,546	993
Clusters (firms)	151	150	151	57
R ²	0.914	0.671	0.581	0.556

(a) Table II: debt effects by lender rating and number of lenders

Table III
Effect of Registry Expansion on ln(Debt), Subsample of Firms with at Least One Rating of 2 before Expansion

This table presents the DD effect of the registry expansion announcement (interim period) and public information (postexpansion period) on how the share of lending by banks that assigned a rating of 2 changes relative to the share by banks that assigned a rating of 1 to the same firm, using specification (6) estimated with OLS:

$$\ln(\text{Debt}_{ijt}) = \alpha_{ij} + \xi_{jt} + \delta_{\text{Treat}_i=1} + \gamma_{\text{Interim}_\text{Bank2}} \text{Treat}_i \times \text{Interim}_t \times \text{BankAssigned2}_j + \gamma_{\text{Post}_\text{Bank2}} \text{Treat}_i \times \text{Post}_t \times \text{BankAssigned2}_j + \epsilon_{ijt}$$

Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Sample: firms with multiple lenders, whose highest (worst) risk rating during the pre-announcement period is a 2, and with at least one rating of 2 (firms with only good ratings excluded). Dependent variables: (log) debt of borrower i in month t with the banks that assigned rating j . Right-hand-side variable of interest: triple interaction between a dummy equal to one if borrower i was in the treatment group, dummies equal to one during the interim and postexpansion periods, and a dummy equal to one if the lender assigned a rating of 2. Coefficients $\gamma_{\text{Interim}_\text{Bank2}}$ and $\gamma_{\text{Post}_\text{Bank2}}$ are the DD estimates. Standard errors are heteroskedasticity robust and clustered at the firm level. ** indicates statistical significance at the 5% level.

Dependent Variable	ln(Debt _{ijt})
Effect on debt by banks that assigned rating of 2: interim period (2 months) (<i>Treat</i> × <i>Interim</i> × <i>BankAssigned2</i>)	-0.214** (0.106)
Effect on debt by banks that assigned rating of 2: postexpansion (12 months) (<i>Treat</i> × <i>Post12</i> × <i>BankAssigned2</i>)	-0.319** (0.158)
Firm × bank fixed effects	Yes
Bank × month dummies	Yes
Treat × month dummies	Yes
No. of observations (firm-month-bank)	4,749
Clusters (firms)	151
R ²	0.921

(b) Table III: triple-difference effect for banks with rating 2

7.7 Table V: Placebo Tests

Read:

- Panel A pretends that the registry expansion occurred in April 1999.
- Panel B pretends that the old information-sharing cutoff was \$300,000.
- Across both placebo designs, the debt and default coefficients are small and statistically insignificant.

Economic message: Table V supports the validity of the design. The main results do not arise mechanically from the sample-selection method, seasonality, or generic changes around debt thresholds.

7.8 Table VI: Firms with Perfect Credit Records

Read:

- Table VI studies firms whose worst pre-expansion rating is 1.
- For multi-lender firms, debt falls by about 5.4% in the postexpansion period, with significance at the 10% level.
- Default effects are not statistically meaningful for these firms.
- Single-lender firms again show no negative debt response.

Economic message: Even firms with initially clean records may borrow less when information becomes public, but the effect is much smaller than for firms with bad news. This is consistent with public information increasing vulnerability to future coordination failures.

Table IV
Effect of Registry Expansion on Default Hazard Rate, Subsample of Firms with at Least One Rating of 2 before Expansion

This table presents the estimated DD effect of the registry expansion announcement (interim period) and public information (postexpansion period) on default hazard rates, using specification (7) estimated with OLS:

$$1|Default_{it} = 1|Default_{it-1} = 0|_{it} = \xi'_i + \lambda_{Interim} \cdot Treat_i \times Interim_t + \lambda_{Post} \cdot Treat_i \times Post_t + \zeta_{it}.$$

Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Sample: firms whose highest (worst) risk rating during the pre-announcement period is a 2, and with at least one rating of 2 (firms with only good ratings excluded). Column 1 (2) is estimated over the subsample of firms with multiple lenders (a single lender) before the expansion announcement. Dependent variable: conditional default of borrower i in month t with any bank. Coefficients $\lambda_{Interim}$ and λ_{Post} are the DD estimates of the effect of the registry expansion announcement and actual expansion on default hazards. Standard errors are heteroskedasticity robust and clustered at the firm level. * indicates statistical significance at the 10% level.

Dependent Variable Subsample	Default Hazard: 1 If Firm in Default at t , and Not at $t-1$	
	Multiple Lenders (1)	Single Lender (2)
Interim Period		
Effect on default hazard rate: interim period (2 months)	0.056* (0.030)	-0.008 (0.028)
Effect on default hazard rate: postexpansion (12 months)	0.030* (0.016)	0.033 (0.022)
Month dummies	Yes	Yes
No. of observations (firm-month)	2,546	993
Clusters (firms)	151	57
R^2	0.104	0.057

(a) Table IV: default hazard effects

Table V
Placebo Tests

This table presents placebo tests that replicate the DD estimations on (log) debt from Tables II and IV over different samples. Panel A. Placebo 1: Estimates where the sample is constructed as if the registry expansion announcement occurred in April 1999, 1 year after actual expansion. Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) between January to March 1999 (instead of 1998). Sample: firms whose highest (worst) risk rating during this is a 2, and with at least one rating of 2 (firms with only good ratings excluded). Panel B. Placebo 2: Estimates where the sample is constructed as if the registry cutoff rule is \$300,000 before the expansion, instead of \$200,000. Treatment (control) firms are those with total debt between \$250,000 and \$300,000 (\$300,000 and \$350,000) during the pre-announcement period (January to March 1998). Sample: firms whose highest (worst) risk rating during the pre-announcement period is a 2, and with at least one rating of 2 (firms with only good ratings excluded). Standard errors are heteroskedasticity robust and clustered at the firm level.

Dependent Variable	$\ln(Debt_{it})$ (1)	$\ln(Debt \text{ from Banks w/Rating} = 2_{it})$ (2)	Default Hazard: 1 if Firm in Default at t , and Not at $t-1$ (3)
Panel A. Placebo 1: Assuming Registry Expansion Occurred in 1999			
Placebo effect: interim period (2 months)	0.020 (0.031)	-0.001 (0.033)	0.007 (0.017)
Placebo effect: postexpansion (12 months)	-0.027 (0.022)	-0.036 (0.055)	0.002 (0.011)
Firm fixed effects, month dummies	Yes	Yes	Yes
No. observations (firm-months)	11,272	10,696	8,081
Clusters (firms)	666	666	659
R^2	0.547	0.925	0.088
Panel B. Placebo 2: Assuming Pre-expansion Debt Cutoff at \$300,000			
Placebo effect: interim period (2 months)	0.034 (0.081)	0.057 (0.076)	-0.017 (0.034)
Placebo effect: postexpansion (12 months)	-0.029 (0.052)	-0.007 (0.122)	0.007 (0.023)
Firm fixed effects, month dummies	Yes	Yes	Yes
In sample after default	Yes	Yes	No
No. of observations (firm-months)	2,258	2,124	1,725
Clusters (firms)	134	134	134
R^2	0.513	0.920	0.103

(b) Table V: placebo tests

7.9 Table VII: Debt Concentration

Read:

- Debt HHI rises after the registry expansion, especially in months 7–12 after the expansion.
- The log number of lenders falls in sign but is not statistically significant.
- The fraction of debt with the top lender rises, indicating that concentration is driven by a larger role for the main creditor.

Economic message: When public information makes inter-lender coordination more powerful, firms respond by concentrating credit relationships. This endogenous change can mitigate future coordination failures.

7.10 Figure A1: Numerical Solution of the Theoretical Framework

Read:

- Panel A shows that making low-creditworthiness information public produces a large expected decline in lending by the bank that holds the bad signal.

- The effect of revealing high-creditworthiness information is much smaller because the prior probability of a good rating is already high.
- Panel B shows that the effect of information sharing on average lending is ambiguous and depends on cost curvature.

Economic message: The model explains why the empirical analysis focuses on bad news. A bad rating has much more informational content than a good rating in a sample where good ratings are very common.

Table VI
Effect of Registry Expansion on Credit Outcomes of Firms with Perfect Credit Records

This table presents estimates of the same DD effects as in Tables II and IV, but over the subsample of firms with perfect credit records before the registry expansion announcement. Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Sample: firms with multiple lenders, whose highest (worst) risk rating during the pre-announcement period is a 1 (firms with a bad rating with any bank excluded). Standard errors are heteroskedasticity robust and clustered at the firm level. * indicates statistical significance at the 10% level.

Dependent Variable Subsample: # Lenders Pre-expansion	ln(Debt _{it})		Default Hazard: 1 if Firm in Default at <i>t</i> , and Not at <i>t</i> - 1	
	Multiple (1)	Single (2)	Multiple (3)	Single (4)
Effect on dependent variable: interim period (2 months)	-0.0455 (0.0591)	0.0468 (0.0472)	0.0026 (0.0047)	-0.0068 (0.0074)
Effect on dependent variable: postexpansion (1 to 12 months)	-0.0543* (0.0281)	0.0550 (0.0604)	0.0033 (0.0038)	-0.0037 (0.0068)
Firm fixed effects	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes
In sample after default?	Yes	Yes	No	No
No. of observations (firm-month)	17,639	9,160	13,370	9,073
Clusters (firms)	891	616	891	616
R ²	0.540	0.512	0.183	0.167

(a) Table VI: firms with perfect credit records

Table VII
Effect of Registry Expansion on Firm Debt Concentration

This table presents the DD effect of the registry expansion announcement (interim period) and public information (postexpansion period) on debt concentration, using specification (5) estimated with OLS:

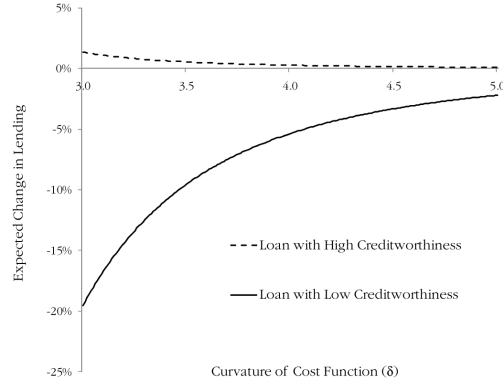
$$y_{it} = \alpha_i + \xi_i + \gamma_{interim} \text{Treat}_i \times \text{Interim}_t + \gamma_{post} \text{Treat}_i \times \text{Post}_t + \varepsilon_{it}$$

Treatment (control) firms are those with total debt between \$150,000 and \$200,000 (\$200,000 and \$250,000) during the pre-announcement period (January to March 1998). Sample: firms that during the pre-announcement period 1) have multiple lenders and 2) whose highest (worst) risk rating during the pre-announcement period is a 2, and with at least one rating of 3 (firms with only good ratings excluded). The dependent variables are the debt HHI, (log) number of lenders, and fraction of debt with the main lender, of firm *i* in month *t*. Right-hand-side variable of interest: interaction between a dummy equal to one if borrower *i* was in the treatment group (information not shared before registry expansion), and dummies equal to one during the interim period and postexpansion period (specifications 2, 4, and 6 include separate dummies for the first and second halves of the postexpansion period). Coefficients $\gamma_{interim}$ and γ_{post} are the DD estimates of the effect of the registry expansion announcement and actual expansion on debt. Standard errors are heteroskedasticity robust and clustered at the firm level. * and ** indicate statistical significance at the 10% and 5% levels, respectively.

Dependent Variable	Debt HHI _{it}		ln(#Lenders _{it})		Fraction by Top Lender _{it}	
	(1)	(2)	(3)	(4)	(5)	(6)
Effect on dependent variable: interim period (2 months)	-0.0021 (0.0043)	-0.0021 (0.0043)	0.0092 (0.0075)	0.0092 (0.0075)	-0.0021 (0.0041)	-0.0027 (0.0040)
Effect on dependent variable: postexpansion (1 to 12 months)	0.0148* (0.0078)	0.0148* (0.0078)	-0.0198 (0.0148)	-0.0198 (0.0148)	0.0122* (0.0074)	0.0122* (0.0074)
Effect on dependent variable: postexpansion (1 to 6 months)		0.0094 (0.0070)		-0.0120 (0.0128)		0.0071 (0.0066)
Effect on dependent variable: postexpansion (7 to 12 months)		0.0194** (0.0092)		-0.0261 (0.0175)		0.0153* (0.0086)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Month dummies	Yes	Yes	Yes	Yes	Yes	Yes
No. of observations (firm-month)	17,577	17,577	17,577	17,577	17,577	17,577
Clusters (firms)	1,042	1,042	1,042	1,042	1,042	1,042
R ²	0.826	0.826	0.784	0.784	0.817	0.817

(b) Table VII: debt concentration effects

Panel A: Expected Change in Lending Conditional on Bank's Private Information



Panel B: Effect of Information Sharing on Average Lending

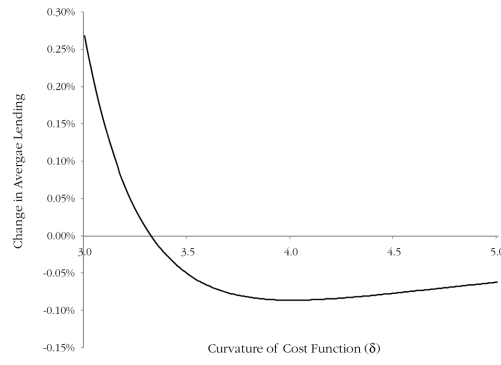


Figure A1. Numerical solution of theoretical framework. All panels are drawn using the following parameter assumptions: $p = 0.9$, $\theta_h = 1$, $\theta_l = 0.25$, and $R = 1$. Panel A plots the publicity multiplier conditional on the creditworthiness (θ_h or θ_l) privately observed by the bank prior to information sharing. For type θ_h and θ_l , this is the expected change in lending as a result of information sharing as a fraction of the level of lending without sharing, defined in the Appendix as $\% \Delta L_h$ and $\% \Delta L_l$, respectively. Panel B plots the unconditional percentage change in the average amount of lending done by each bank as a result of information sharing, defined in the Appendix as $\% \Delta L_{average}$.

Figure A1: numerical solution of the theoretical framework

8 Short summary

- The paper studies the 1998 expansion of Argentina's public credit registry.
- The reform made previously private borrower ratings public for many smaller borrowers.
- The delay between announcement and implementation creates an interim period that isolates the effect of anticipated publicity from the effect of receiving new information.
- Treatment firms are just below the old \$200,000 information-sharing threshold; control firms are just above it.
- Banks that had privately assigned a rating of 2 cut lending immediately after announcement when the borrower has multiple lenders.
- Banks with good ratings do not cut until after other banks' bad ratings become public.
- Single-lender firms are not affected, which supports the coordination interpretation.

- Default hazards rise for affected multi-lender firms during the interim period.
- Placebo tests do not reproduce the results.
- In the longer run, firms concentrate borrowing with fewer or more dominant lenders.

9 Conclusion

9.1 Core conclusion

Hertzberg, Liberti, and Paravisini provide direct evidence that creditors to the same firm have incentives to coordinate. When a lender learns that its negative private assessment of a borrower will become public, it reduces lending before it receives any new information from other lenders. This response is strongest for multi-lender firms and absent for single-lender firms.

9.2 Economic intuition

The intuition is that a lender’s payoff depends on what other lenders do. If one bank expects other banks to withdraw after seeing bad news, it may withdraw first. Public information is therefore more powerful than private information because it forecasts other lenders’ actions. This is the publicity multiplier.

9.3 Incremental contribution revisited

The paper’s distinctive contribution is the clean separation of information from publicity. In many credit-market events, it is hard to tell whether lenders withdraw because they learned that fundamentals are worse or because they expect other lenders to withdraw. The registry expansion’s interim period solves this problem: banks have no new external information, but they know their own information will become public.

9.4 Implications for disclosure policy

The findings imply that transparency can have costs in environments with strategic complementarities. Public credit registries may reduce asymmetric information, but they may also make creditor runs more likely when borrowers are close to distress. Policymakers should distinguish between ordinary information benefits and the coordination externalities created when many creditors react to the same public signal.

9.5 Limitations

The design is local to borrowers around Argentina’s old \$200,000 registry threshold in 1998. The sample focuses on relatively small borrowers with formal bank credit. The paper identifies lender behavior around a specific information-sharing reform, not the full welfare effect of credit registries in all settings. Some selection around the threshold is possible, although the paper directly addresses this with parallel-trend evidence and placebo tests.

9.6 Takeaway

The paper’s enduring message is that public information can coordinate actions. In credit markets, this may be dangerous when borrowers rely on several lenders and are close to distress. The same

information can have a much larger effect when it is public than when it is private, because each lender reacts not only to the signal but also to what the signal will make other lenders do.